

Gemtuzumab ozogamicin (Mylotarg®)

Gemtuzumab ozogamicin (Mylotarg®) is a cancer drug. It is used to treat acute myeloid leukaemia (AML).

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What is gemtuzumab ozogamicin (Mylotarg®)?

Gemtuzumab ozogamicin is also called Mylotarg®. Gemtuzumab ozogamicin is used to treat [acute myeloid leukaemia](#).

Gemtuzumab ozogamicin is an [antibody drug conjugate](#). This means it has 2 parts:

- gemtuzumab, which is a type of targeted therapy drug called a monoclonal antibody.
- ozogamicin, which is a chemotherapy drug.

In this information, we use the term gemtuzumab to mean gemtuzumab ozogamicin.

Gemtuzumab targets a protein called CD33. CD33 is found on the surface of some leukaemia cells and some types of developing white blood cell. Gemtuzumab locks on to the CD33 protein. It releases ozogamicin into the leukaemia cells. This damages and kills them.

It is best to read this information with our general information about [targeted therapy drugs](#), [chemotherapy](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How gemtuzumab is given

You usually have gemtuzumab during a stay in hospital. It may sometimes be given in a day unit as an outpatient.

You usually have gemtuzumab with chemotherapy drugs. It can be given in combination with chemotherapy drugs called [daunorubicin and cytarabine](#).

We have separate information if you are having [gemtuzumab, with daunorubicin and cytarabine](#).

During a course of treatment, you will meet someone from your cancer team, such as a:

- blood cancer doctor (haematologist)
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Your doctor, nurse or pharmacist will discuss your treatment plan with you. They will explain when you will be given gemtuzumab and your chemotherapy drugs. They will explain how many treatments you will have. They may give you a copy of the treatment plan to keep.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your targeted therapy.

You have gemtuzumab as a drip into a vein (intravenous infusion). The drip will usually run through a pump. This will give you the gemtuzumab over 2 hours.

You have the drip through 1 of the following:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in the arm or hand
- a [central line](#) – a fine tube that goes under the skin of the chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are being given this treatment or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Before treatment, you will have medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick

- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Changes in blood pressure

Gemtuzumab can affect your blood pressure during and after your infusion. Your blood pressure will be checked during this time.

Your blood pressure may become higher or lower than normal. Tell your nurse or doctor if you have headaches or feel dizzy. These can be signs of blood pressure changes.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness

- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Effects on the liver

This treatment can affect how the liver works. You will have regular blood tests to check this. Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
-

unexplained bleeding or bruising

- pain in the right side of your tummy (abdomen)
- weight gain
- swollen face, ankles or legs.

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Feeling sick

The nurses will give you anti-sickness drugs regularly. This will help prevent or control sickness during your treatment in hospital. If you [feel sick or are sick \(vomit\)](#), tell your nurse or doctor straight away. They can change the anti-sickness treatment. Or they might change the way you have it so that it works better for you.

If you are at home, contact the hospital on the 24-hour number as soon as possible.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

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- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Feeling tired (fatigue)

This treatment can make you [feel very tired](#) and you will need a lot of rest. You may get tired easily for some months after treatment ends. Gentle exercise, like short walks, can help you feel less tired. The tiredness will slowly get better.

If you feel sleepy, do not drive or use machinery.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time

- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS) during this treatment. When treatment makes large numbers of leukaemia cells die and break down quickly, they release lots of waste products into the blood. This can affect how well the kidneys work. It may also cause problems with the heart rhythm.

You may have a higher risk of TLS if your white blood cell count is very high before starting treatment. Your doctor may reduce your white blood cell count before you start treatment by giving you:

- a drug called [hydroxycarbamide](#)
- treatment called [leukapheresis](#).

You will have regular blood tests to check for TLS. If you are at risk of TLS, you may have:

- extra fluids through a drip
- medicines such as [rasburicase](#) through a drip, or allopurinol as tablets.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain

- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex in the first few days after treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert health professionals and people living with cancer.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
-

explain medical words

- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

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References



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[Cancer Information Development team](#)

Author

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help